CCS3104-Group Project

IKEA AR Interior Design Assistant

With Community-based Design Reuse

Focus: Integrating AR, AI, and Community

Group Member

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"Transforming furniture shopping from a static catalog into an interactive, intelligent, and socially supported design process."

Company Overview: IKEA’s Legacy and Digital Evolution

Founding & Philosophy

Founded in 1943 by Ingvar Kamprad; built on "Democratic Design" balancing Form, Function, Quality, Sustainability, and Low Price.

Core Innovation

Pioneered flat-pack furniture in the 1950s, revolutionizing affordability and global logistics by reducing transportation costs.

Global Reach & Digital Strategy

Operates in 60+ markets. Currently investing heavily in e-commerce, smart home solutions, and mobile apps to integrate technology with daily living.

Organizational Structure: Strategic Project Alignment

Inter IKEA Group

Brand Ownership

Oversees the IKEA concept and global brand identity.

Product Development

Provides core furniture data and design direction.

Ingka Group

Retail Operations

Manages physical stores and customer touchpoints.

Digital Services

Primary host for the AR Assistant's deployment and user interface.

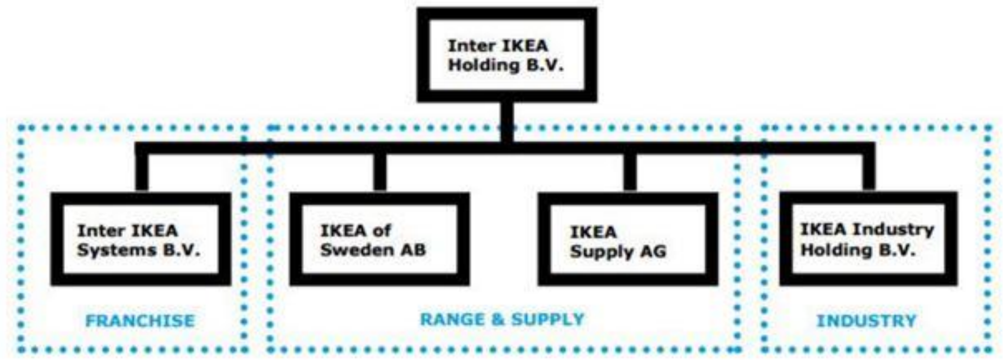
Key Departments

Digital Innovation & IT

Lead development of AR and AI modules.

Marketing & CX

Drive user adoption and community engagement.



Strategic Integration

The AR Assistant bridges the gap between product design (Inter IKEA) and retail experience (Ingka), ensuring a seamless digital-to-physical customer journey.

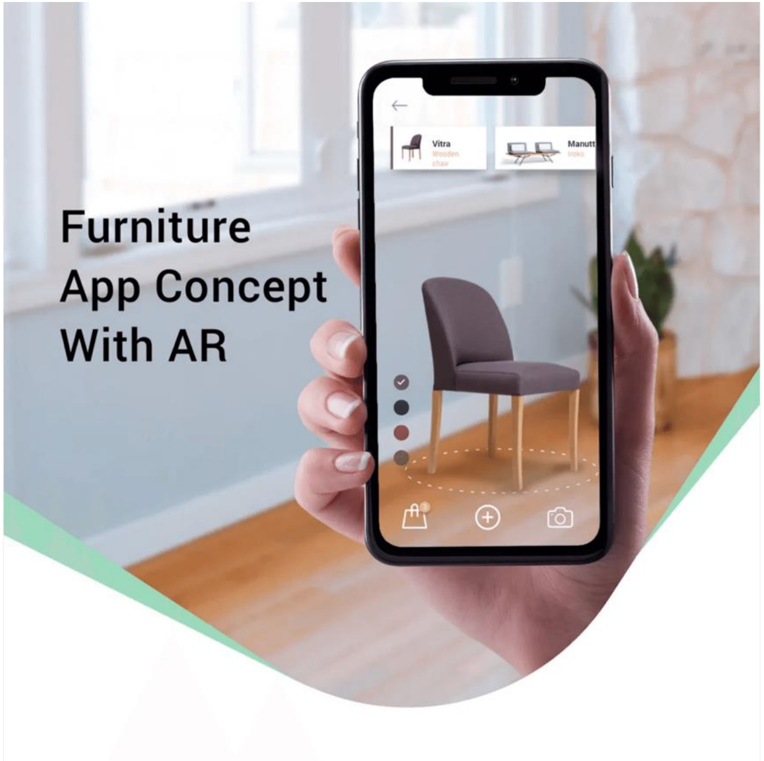
The Problem & Solution: Bridging the Gap in Interior Design

The Challenge

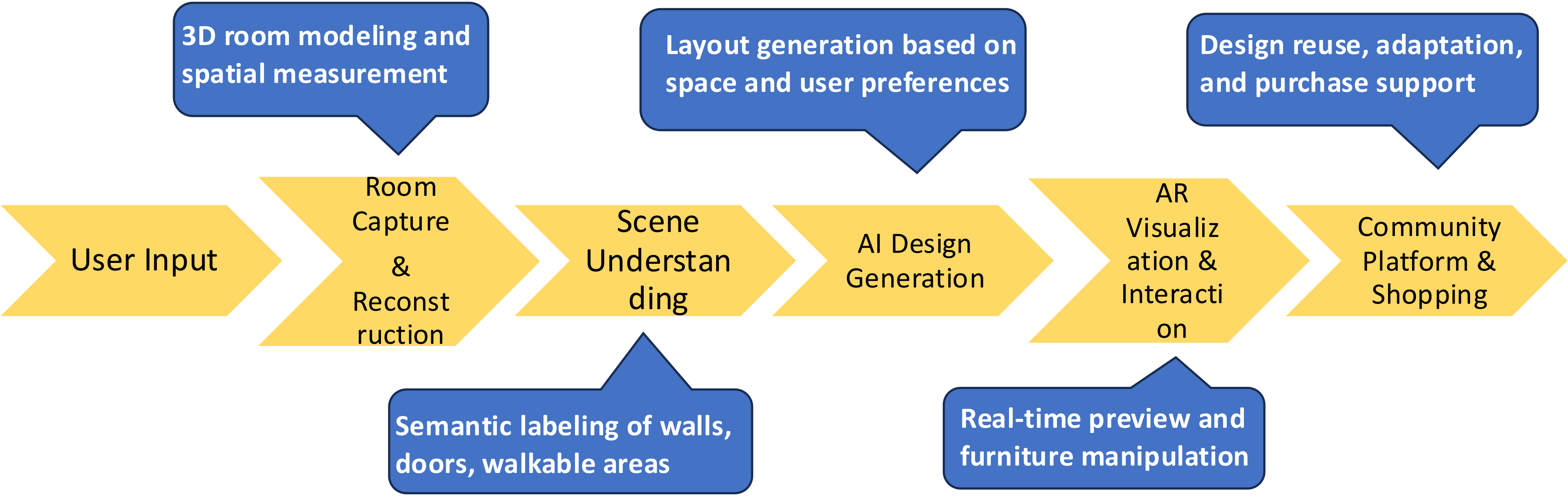
- Inaccurate spatial understanding and measurement errors.
- Uncertainty regarding furniture size compatibility and style fit.
- Budget overruns due to trial-and-error purchasing.

The Solution

- *Intelligent AR Assistant:*
  A mobile platform using Computer Vision and Generative AI.
- Seamless Visualization:
  Bridges the gap between imagination and physical space.
- *Social Validation:*
  Combines AI precision with community-driven design wisdom.



Architecture （System Architecture）



“This architecture supports an end-to-end workflow from room scanning to purchase decision.”

Core Features (Part 1): Room Scanning & AI Layout Generation

Room Capture & Reconstruction

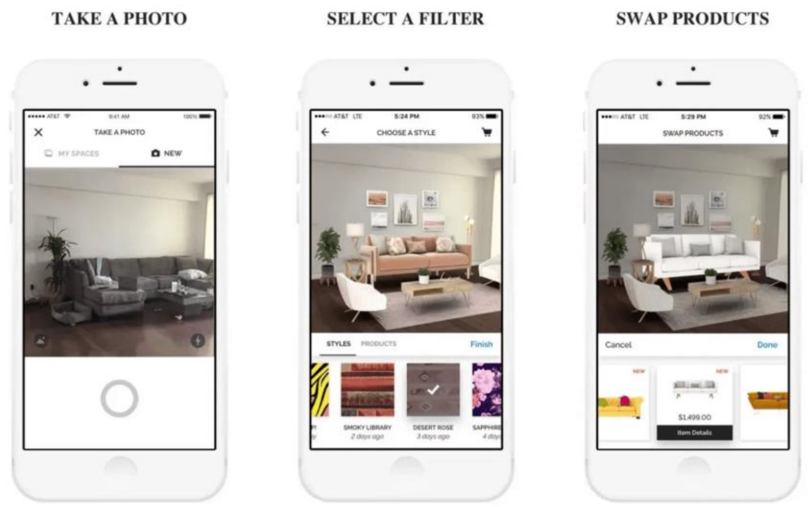
Utilizes smartphone cameras to detect boundaries, measure dimensions, and create high-fidelity 3D spatial models of the user's environment.

Semantic Scene Understanding

Identifies existing furniture, walkable areas, and clearance zones (e.g., door swings) to ensure all generated designs are physically feasible.

AI-Assisted Generation

Processes natural language inputs (e.g., "Scandinavian style, budget RM3000") to produce multiple optimized layout proposals with conflict detection.



Core Features (Part 2): AR Visualization & Community Reuse

Immersive AR Interaction

- **Real-time Visualization:**

  Overlays virtual furniture with light estimation and occlusion handling for high realism.
- **Interactive Customization:**

  Drag, rotate, and snap furniture to walls/floors while maintaining spatial correctness.

Socially-Powered Design

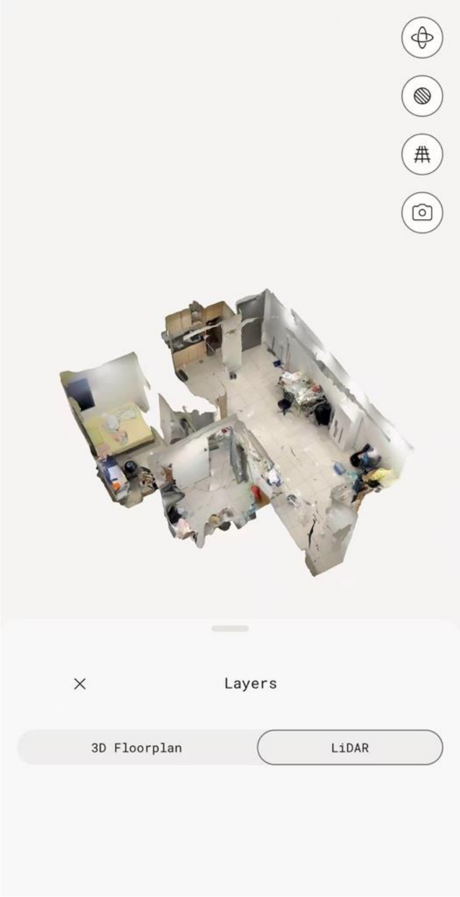
- **Community Platform:**

  Browse and adapt designs from users with similar room layouts and furniture preferences.
- **Automatic Adaptation:**

  System scales and adjusts community designs to fit the user's specific room dimensions.

GUI Prototype

https://www.figma.com/make/Dk1Pk3F2yo4HSRjtWzaVfi/Mobile-GUI-Design-Report--Copy-?t=ZYRziibj5U8PkEbv-1



GUI Prototype https://camel-secure-21405066.figma.site

Justify & Compare

Aspect	Traditional Offline Shopping	Traditional AR Apps	Proposed 3D AR-based Solution
Measurement Accuracy	Manual measurement, prone to human error	Relies on estimated dimensions	Accurate 3D room reconstruction
Spatial Constraints	Often ignored until purchase	Limited awareness of real constraints	Explicit modeling of walkways and clearances
Visualization	Based on imagination or static display	2D screen-based AR overlay	Real-time AR in reconstructed 3D space
Layout Feasibility	High risk of blocking paths or doors	Errors amplified by incorrect scale	Automatic detection of layout conflicts
Time & Effort	Time-consuming and physically demanding	Requires repeated trial-and-error	Efficient, guided design process
Decision Confidence	Low, relies on experience	Medium, but error-prone	High, supported by simulation and validation

Targeted Customers: Addressing Diverse User Needs

Renters Small-Apartment Residents

- Focus on size compatibility and space optimization.
- Flexible, non-permanent solutions for temporary living.
- Avoid costly mistakes by previewing fit before purchase.

First-time Homeowners

- Cohesive style guidance for complete room furnishing.
- Confidence in large-scale investments and furniture sets.
- Professional-grade visualization for empty spaces.

Busy Working Professionals

- "One-click" design generation to save time and effort.
- Automated shopping lists for efficient purchasing.
- Explainable AI recommendations for quick decisions.

Empowering non-experts to create functional and aesthetically pleasing living spaces with intelligent, user-centric tools.

Promotion Strategies (Part 1): Digital & In-store Integration

Seamless App Integration

Embed the AR assistant directly into the existing IKEA mobile app to leverage the current global user base and reduce friction.

Omnichannel Promotion

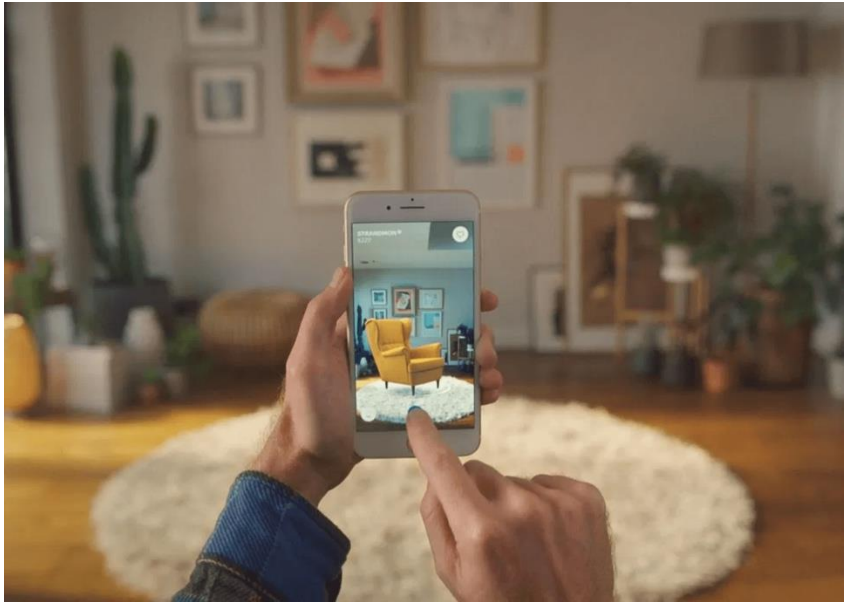
Cross-promote via the IKEA website and online store to ensure a consistent digital journey from browsing to visualization.

In-store AR Demo Zones

Dedicated areas in physical stores where customers can experience the technology firsthand with staff guidance.

QR Code Accessibility

Instant access via QR codes on product tags, allowing customers to "try before they buy" in their own homes.



Promotion Strategies (Part 2): Community & Sustainability

Community Engagement

Community-Driven Growth

Highlight popular designs and reward top contributors with badges to foster an active, self-sustaining design ecosystem.

Social Sharing

Encourage users to share their AR-designed spaces on social media, creating organic brand advocacy and viral reach.

Eco-Consciousness

Sustainability Marketing

Promote eco-friendly options and "Sustainability Scores" for each design proposal to guide responsible consumption.

Brand Alignment

Position the service as a tool for responsible living, aligning with IKEA’s "People & Planet Positive" global strategy.

"Driving Engagement through Social Validation and Eco-Consciousness."

Social & Economic Impact: Creating Value for All

Social Impact

- **Democratizing Design**
  Lowers the barrier to professional interior design, empowering non-experts to create functional spaces.
- **Community Building**
  Fosters collaboration and shared learning through the design reuse platform and social interaction.
- **Stress Reduction**
  Enhances user confidence and reduces the anxiety associated with complex furniture selection processes.

Economic Value

- **Increased Conversion**
  Drives sales by providing high-fidelity previews that reduce purchase hesitation and decision friction.
- **Operational Efficiency**
  Significantly reduces after-sales costs by minimizing product returns due to size or style mismatches.
- **Data-Driven Insights**
  Generates valuable data on customer preferences and space usage to optimize future product development.

"Enhancing the customer journey while driving sustainable business growth through digital innovation."

Environmental Impact: Designing for a Greener Future

Minimized Returns

Accurate measurements and AR previews significantly reduce the carbon footprint associated with product returns and reverse logistics.

Eco-Friendly Guidance

The AI system prioritizes sustainable materials and energy-efficient layouts in its design recommendations.

Waste Reduction

Prevents the purchase of incompatible items that might otherwise end up as waste, promoting a more circular economy.

Data-Driven Sustainability

Provides insights into space usage patterns to optimize future product development for smaller, greener homes.

Aligning digital innovation with IKEA's "People & Planet Positive" strategy to foster responsible consumption and sustainable living.

Conclusion: The Future of Intelligent Home Furnishing

A Comprehensive Ecosystem

The IKEA AR Assistant is more than a tool; it is a holistic ecosystem that combines cutting-edge technology with community wisdom to empower every user.

Future Outlook

Potential for deeper integration with smart home IoT and advanced generative design for fully autonomous, personalized interior planning.

"Innovation that balances form, function, and sustainability for the digital age, creating a better everyday life at home."

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Innovation that balances form, function, and sustainability.

Thanks

“ Bringing Imagination to Life.”